HAOYU WANG

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Profile

Haoyu (PhD Candidate, Financial Engineering, University of Auckland) builds and validates quantitative models and uses AI – LLMs and AI-assisted (vibe) coding – to automate data, reconciliation, and reporting tasks, turning hours of manual work into fast, repeatable pipelines.

Publications

Optimizing Stock Market Prediction and Stock Trading Strategies with Deep Learning Models Enhanced by Nonlinear Feature Identification and Robust Prediction Evaluation 2026
Financial Innovation, 12, 117. DOI: 10.1186/s40854-026-00929-6

Information-aware Valuation and Dynamic Pricing for Textual Data in Mobile Environments 2026
Information Processing and Management, 63 (7), 104855. DOI: 10.1016/j.ipm.2026.104855

Can Ensemble Learning Reveal Real Estate Value – A Study of England and Wales Cases Anonymous 2026
with Yinze Ji and Yishuang Xu. *Journal of Chinese Economic and Business Studies*, 1–36. Routledge. DOI: 10.1080/14765284.2026.2722767

Optimal Profit-Making Strategies in Stock Market with Algorithmic Trading 2024
Quantitative Finance and Economics, 8 (3), 546. American Institute of Mathematical Sciences (AIMS).

Entropy-Regularized Portfolio Optimization with Reinforcement Learning Control 2025
Working Paper, SSRN: abstract_id=5276183

AI & Automation

Apply LLMs (Claude, Codex) and vibe coding to rapidly prototype tools and automate workflows – data cleaning, reconciliation, report generation, and research pipelines – turning multi-hour manual tasks into repeatable scripts.

Skilled in prompt engineering and AI-assisted development to accelerate model building and analysis, raising throughput without sacrificing accuracy or compliance.

Work / Project Experience

University of Auckland 2025 — Present
Finance Researcher — Finance (Investment) & Fintech *Auckland, New Zealand*

- Research and construct AI-platform cross-sectional multi-factor models/portfolio optimisation with machine-learning/DL trading strategy, achieving a high Sharpe ratio and lower max drawdown in the US market.

Oxford Finance – Turner NZ Oct 2025 — Jan 2026
Finance Analyst (Fixed-Term Project) *Auckland, New Zealand*

The Chinese University of Hong Kong, Shenzhen 2024
Research Assistant *Shenzhen, China*

Xi'an Jiaotong-Liverpool University 2023 — 2024
Research Assistant, School of Mathematics and Physics *Suzhou, China*

S&P Global (IHS Markit) Oct 2023 — Feb 2024
Quantitative Research Intern *Shanghai, China*

Education

University of Auckland Dec 2024 — Present
PhD, Financial Engineering — Full Scholarship *Auckland, New Zealand*

Xi'an Jiaotong-Liverpool University Sep 2022 — Apr 2024
MSc, Financial Mathematics

Chongqing Finance and Economics University Sep 2018 — Jun 2022
BSc, Finance

Selected Projects & Research

Hybrid RL Market Making with Avellaneda–Stoikov Backbone 2026

- Order-book-aware high-frequency market making combining reinforcement learning with an Avellaneda–Stoikov backbone, tested on real Binance Futures BTCUSDT L2 data.

NZ Electricity Market Spot-Price Forecasting 2025

- End-to-end forecasting pipeline for NZEM half-hourly spot prices, built on real Electricity Authority/EMI data.

Achievements

Citadel Trading Invitational (Australia, 2025): competed across market-making simulation, quant simulation, and a timed coding-challenge track.